Thermal Biomarker Analysis for Diabetic Foot: A Confounder-Controlled Study

Jayashree J and Lavanya Singh
School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology

1. Introduction and Clinical Context

Diabetic foot ulcers represent one of the most debilitating and costly complications associated with diabetes mellitus. Each year, approximately 18.6 million individuals develop a diabetic foot ulcer, which often precedes the majority of lower-extremity amputations in diabetic patients. Furthermore, the five-year mortality rate following a major amputation surpasses that of several prevalent cancers. The likelihood of recurrence is nearly forty percent within one year post-healing. Consequently, the clinical significance of any screening technology is rooted in its ability to identify a foot at risk before tissue deterioration occurs, rather than in categorizing a wound that has already developed. Infrared thermography has garnered ongoing interest as a potential screening tool. A thermogram is non-invasive, does not involve ionizing radiation, is cost-effective to obtain, and can now be captured using smartphone attachments that have been formally evaluated against clinical-grade cameras. The introduction of annotated public benchmarks, particularly the INAOE Plantar Thermogram Database, has facilitated rapid advancements in automated analysis, resulting in a substantial body of literature that reports classifiers distinguishing diabetic subjects from control individuals with accuracies ranging from 0.90 to 0.99. This study poses a preliminary question that has largely been overlooked in existing literature. In the benchmark utilized by most of these studies, the control participants are typically younger, while the diabetic patients are older, with an average age difference of 28.2 years. The prevalence of diabetes increases significantly with age. Therefore, a model that has access to age or any age-correlated feature can achieve high apparent accuracy without deriving any thermal-physiological information. In such cases, the reported metric reflects the demographic composition of the recruited cohort rather than the actual efficacy of the screening modality.

Central question. How much of the accuracy reported on this benchmark is attributable to thermal physiology, and how much to the age structure of the cohort? The study does not propose a classifier. It audits the dataset.

---

2. The Dataset

2.1 Source and provenance

This report's analyses utilize the INAOE Plantar Thermogram Database, which was released by the Instituto Nacional de Astrofísica, Óptica y Electrónica in Mexico and is publicly available for research purposes. This database was specifically created to facilitate the investigation of diabetic foot complications and has established itself as the primary reference dataset in this area; a significant portion of the published machine-learning research on plantar thermography is based on it.

The complete release consists of 334 plantar thermograms, representing both feet of 167 individuals, of which 122 have been diagnosed with diabetes mellitus, while 45 serve as non-diabetic controls. The images were captured following a standardized protocol as outlined by the database authors and are accompanied by derived temperature files that categorize each foot into anatomically defined regions. The dataset is de-identified at the source, thus no further ethical approval was necessary for the current secondary analysis.

2.2 What this study uses

The database is available in two formats: the thermal images and tabular records that provide the average temperature for each anatomical region, along with essential demographic and anthropometric data. This study focuses solely on the tabular records. No image processing, segmentation, or deep feature extraction is conducted.

This decision is intentional and stems from the research question. The aim of the study is not to determine whether a more advanced representation could yield additional insights, but rather to ascertain if the discrimination observed in this benchmark is due to thermal physiology or the composition of the cohort. This question is most effectively addressed through the regional summaries, where each feature has a physiological significance, and no representational choice can be held accountable for a null result. The associated limitation is that aspects such as spatial texture, gradient structure, and localized hot-spot morphology are entirely excluded from the analysis.

2.3 Cohort composition

Following the processes of parsing and type coercion, the analysis cohort consists of 167 subjects. The ratio of classes is roughly one control subject for every 2.7 diabetic subjects. Consequently, all classifiers were set up with balanced class weights, and macro-averaged metrics were consistently employed to ensure that the smaller control group would not be overlooked by the model.

Table 1. Composition of the analysis cohort.

Property	Control (CG)	Diabetic (DM)	Total
Subjects	45	122	167
Proportion of cohort	26.9 per cent	73.1 per cent	100 per cent
Thermograms contributed	90	244	334
Mean age	27.8 years	56.0 years	—
Age standard deviation	8.2 years	10.6 years	—
Age range represented	Predominantly 21–40	Predominantly 44–84	21–84

2.4 Variables in the tabular records

Each row of the source file corresponds to one subject and carries an identifier prefixed CG or DM, three anthropometric fields, age, gender, and twelve regional temperature values, six for each foot.

Table 2. Fields available in the distributed tabular records.

Field	Description	Used in this study
Subject	Identifier prefixed CG for controls and DM for diabetic subjects	Yes, to derive the class label
Gender	Recorded sex of the subject	No
Age	Age in years at acquisition	Yes, as a clinical covariate and as the confounder under study
Weight, Height	Anthropometric measurements	Indirectly, through body mass index
IMC	Body mass index, from the Spanish índice de masa corporal	Yes, as the second clinical covariate
R_General, L_General	Whole-foot mean temperature, right and left	Yes, in the thermal block
R_LCA … R_TCI	Mean temperature of each of the five angiosome regions, right foot	Yes, in the thermal block
L_LCA … L_TCI	Mean temperature of each of the five angiosome regions, left foot	Yes, in the thermal block

2.5 The angiosome regions

The regional partition follows the angiosome concept, which divides tissue into territories according to the artery supplying each. This matters for the present analysis because it means the regions are not arbitrary spatial tiles: each corresponds to a distinct vascular supply, so regional temperature differences carry a specific circulatory interpretation.

Table 3. Plantar regions provided for each foot.

Code	Region	Anatomical location
General	Whole-foot average	Mean across the entire plantar surface
LCA	Lateral calcaneal	Outer aspect of the heel
LPA	Lateral plantar	Outer aspect of the mid and forefoot
MCA	Medial calcaneal	Inner aspect of the heel
MPA	Medial plantar	Inner aspect of the mid and forefoot
TCI	Tibial–calcaneal interface	Region adjoining the ankle

2.6 Preparation of the analysis file

The source workbook contains two groups organized on separate sheets. Both groups were read with the header rows omitted, columns were renamed to adhere to a common schema, and the two data frames were merged. Rows with subject identifiers that did not conform to the expected CG or DM pattern were eliminated, effectively removing trailing notes and blank rows found in the distributed file. Subsequently, all numeric fields were coerced, with any entries that could not be parsed designated as missing rather than being silently defaulted to zero.

The body mass index had a limited number of missing values. These were imputed using the median, which was calculated within the training partition of each cross-validation fold and then applied to the held-out fold, ensuring that the median was never computed across the entire dataset. This aspect is significant: imputing from the complete dataset would result in information leakage from the evaluation data into the training process, thereby inflating all subsequent performance metrics reported. The group label was encoded such that controls were represented as zero and diabetic subjects as one.

Note on data handling. Every preprocessing step that estimates a quantity from data — imputation, standardisation and feature scaling — is implemented as a stage of an estimator pipeline, so that it is re-fitted within each training partition. No statistic derived from a held-out fold is ever used in constructing the model evaluated on it.

2.7 What the dataset does not contain

Three absences limit the inquiries that can be made regarding this data. These limitations are outlined here, along with a description of the dataset, as they influence the overall study design.

Firstly, there is a lack of clinical stratification within the diabetic group. Factors such as the duration of diabetes, glycaemic control, grading of neuropathy, and vascular status are not documented. Consequently, the diabetic group is heterogeneous concerning the very pathology that thermal features are believed to reflect, which may diminish any authentic signal.

Secondly, the dataset is cross-sectional in nature. Each participant is represented only once, at a single point in time. Therefore, no conclusions can be drawn regarding ulcer prediction, progression, or longitudinal changes in thermal readings, and the only outcome available — current diabetes status — serves merely as a proxy for the clinically significant issue of potential future ulceration.

Lastly, and most critically for this study, the two groups were not matched based on age at the time of recruitment. The control volunteers are significantly younger than the diabetic patients. Section 4.2 provides a quantification of this imbalance, and the subsequent sections of the report explore the implications of this disparity.

Why this dataset, given that limitation. The age imbalance is not a reason to avoid the benchmark; it is the reason to audit it. Because so much of the published literature reports performance on this dataset, establishing what those numbers can and cannot mean is more useful than adding another classifier to the same pile.

3. Methodology

3.1 Feature blocks

Three blocks of information were defined so that their contributions could be compared directly.

Block	Contents	Count
Clinical	Age and body mass index. No thermal information.	2
Thermal	Per region: contralateral absolute difference and bilateral mean.	12
Biomarkers	Five physiologically motivated indices derived from the regional temperatures.	5

Design rationale. The clinical block contains no thermal information whatsoever. It is the comparator against which every thermal claim is judged. If a two-variable demographic model matches a nineteen-variable model containing all the thermal data, the modality is not contributing.

3.2 The thermal biomarkers

With R and L denoting right- and left-foot temperatures of a region, and the regional set comprising LCA, LPA, MCA, MPA and TCI:

$$TAI = \text{mean over regions of } |R - L| \quad (1)$$

$$RHHI = \tfrac{1}{2} \cdot [\, \text{SD across regions of right foot} + \text{SD across regions of left foot} \,] \quad (2)$$

$$PTVI = \max - \min, \text{ taken over all ten regional values of both feet} \quad (3)$$

$$ASI = \text{mean absolute deviation of each region from its own foot's mean} \quad (4)$$

$$TSS = 1 - TAI / [\, \tfrac{1}{2} (\text{mean right} + \text{mean left}) + \varepsilon \,], \quad \varepsilon = 1e{-}8 \quad (5)$$

Biomarker	Quantity measured	Physiological rationale
TAI	Left-right difference averaged across regions	Both feet share systemic drivers, so a contralateral difference isolates local pathology

Biomarker	Quantity measured	Physiological rationale
RHHI	Temperature variation between regions within one foot	Uniform perfusion should give a smooth field; patchiness suggests regional compromise
PTVI	Range between hottest and coldest region	Captures localised hot or cold spots that averages conceal
ASI	Deviation of each region from its own foot mean	Per-foot form of spatial heterogeneity
TSS	Rescaled symmetry, higher meaning more symmetric	Mirror of asymmetry, normalised by overall foot temperature

3.3 Nested cross-validation

A model assessed on the data utilized for its selection will invariably seem more effective than it truly is. Consequently, the protocol positions each selection decision within an inner loop that does not access the evaluation data.

The outer loop segments the 167 subjects into five stratified groups, training on four while evaluating on the fifth, rotating through all five; this entire process is repeated ten times with varying partitions, resulting in fifty untouched outer folds. Within each outer training partition, a five-fold inner loop determines hyperparameters, ensemble membership, voting weights, and, when applicable, the decision threshold. The outer fold is accessed only once to document the score.

The outer split list was created once and reused across all feature blocks, ensuring that block comparisons are matched at the fold level and that differences are not influenced by varying partitions. The decision threshold was consistently set at 0.5 for all primary results; a sensitivity analysis utilizing inner-loop threshold selection is detailed in Section 4.7.

3.4 Inference on paired differences

The training sets exhibit substantial overlap across folds, resulting in a positive correlation of fold-level scores. Consequently, a standard paired t-test rejects the null hypothesis far more often than it should. To assess the differences between blocks, the Nadeau-Bengio corrected statistic was employed, which increases the variance term by a factor that corresponds to the ratio of the test set size to the training set size:

$$t = \bar{d} \,/\, \mathrm{sqrt}[\, s^2 \cdot (1/n + n\_test \,/\, n\_train) \,] \qquad (6)$$

where $\bar{d}$ and s^2 represent the mean and variance of the fold-wise differences. Bootstrap percentile intervals are provided as well. The agreement at the individual subject level was assessed using the exact McNemar test on the discordant pairs, which poses a different and more rigorous question regarding whether the inclusion of thermal information alters the classification accuracy of subjects.

3.5 Biomarker validation

Each biomarker underwent validation on two occasions. The unadjusted comparison employed the Mann-Whitney U test, utilizing Cliff’s delta and bootstrap intervals derived from 2000 resamples.

In the adjusted comparison, standardised age, standardised body mass index, and the standardised biomarker were incorporated into a logistic model. Ordinary maximum likelihood is not applicable in this context, as age nearly separates the two groups. Under separation the fitted coefficient and its standard error both diverge, resulting in a loss of power for the usual test precisely at the point where the association is strongest.Firth's method addresses this issue by applying a penalty to the log-likelihood, incorporating one half of the log-determinant of the Fisher information, thereby ensuring that the estimates remain finite.

$$\ell^*(\beta) = \ell(\beta) + \tfrac{1}{2} \cdot \log | I(\beta) | \qquad (7)$$

The biomarker was subsequently evaluated using a penalised likelihood-ratio test instead of a Wald test, as the latter is deemed unreliable in this context. Holm's step-down procedure was employed to manage the family-wise error rate across the five markers, while a nested test with five degrees of freedom assessed the entire panel against a model that included only age and body mass index.

3.6 The four confounder-adjustment designs

No single design is trustworthy when overlap is poor, so four were applied and their disagreements reported rather than reconciled.

Design	Principle	Theoretical cost
Age restriction	Retain only subjects inside the overlapping age band and re-execute the full pipeline	A residual gap remains inside the band, so the estimate is a lower bound
Repeated matching	Pair subjects on the propensity logit within a caliper; repeat over 1000 random orderings	Discards unmatched subjects; only 20 of 45 controls could be paired
Overlap weighting	Retain all subjects, weighting by probability of the opposite group	Balances the propensity score, not necessarily age itself
Stabilised IPTW	Weight by the reciprocal of the propensity, trimmed at the 99th percentile	Unstable under near-separation; reported as a sensitivity check only

The propensity score utilized in the last three designs represents the probability of belonging to a specific group based on age and body mass index. This score is estimated through logistic regression and is constrained within the range of 0.02 to 0.98. Writing e(z) for this score, overlap weights are allocated as

$$w = 1 - e(z) \text{ for the diabetic group;} \quad w = e(z) \text{ for the control group} \qquad (8)$$

This is constrained within the unit interval, ensuring that no individual observation can exert undue influence, and it is designed to achieve exact balance in the propensity score. The balance was confirmed through verification rather than mere assumption, utilizing the standardized mean difference.

$$SMD(v) = (\bar{v}_1 - \bar{v}_0) / \mathrm{sqrt}[(s^2_1 + s^2_0) / 2] \qquad (9)$$

The evaluation was conducted both prior to and following each design, with the standard threshold of 0.1 indicating minimal imbalance.

Greedy matching is influenced by the sequence in which control subjects are handled; therefore, the process was repeated across 1000 random permutations, and the resulting distribution was analyzed instead of relying on a single realization.

3.7 Attribution and integrity check

SHAP values were derived from a random forest comprising 400 trees solely for the purpose of interpretation, in two distinct configurations: one including clinical variables and the other excluding them, the latter being necessary as age tends to completely dominate the attribution. Attribution pertains to a fitted model rather than a population and should never be regarded as evidence of association.

Prior to any modeling, a depth-three decision tree was applied to the raw regional temperatures. If it had achieved perfect accuracy, the labels would be deterministically retrievable from the features, rendering the benchmark ineffective. This verification is cost-effective and is advised as a standard procedure before analyzing any public dataset.

4. Results

4.1 Benchmark integrity

The depth-three tree achieved a resubstituting accuracy of 0.856, which is significantly below one. The class labels cannot be retrieved as a simple deterministic rule from the provided features, and the benchmark is sound in that respect.

4.2 Cohort structure and the extent of the confound

The average age of controls is 27.8 years, while that of diabetic subjects is 56.0 years, resulting in a difference of 28.2 years. Cliff's delta of +0.938 indicates a near-complete stochastic dominance, as approximately 97 percent of the control-diabetic pairings show that the diabetic subject is older. The propensity model yields mean scores of 0.198 for controls and 0.921 for diabetics, with only 49 out of 167 subjects located within the common-support region.

Table 4. Cohort characteristics and separation on age.

Characteristic	Control (CG)	Diabetic (DM)	Separation
Number of subjects	45	122	167 total
Age, mean ± SD (years)	27.8 ± 8.2	56.0 ± 10.6	Difference 28.2 y
Mann-Whitney test	—	—	$p = 1.52 \times 10^{-20}$
Cliff's delta	—	—	+0.938
Propensity score, mean	0.198	0.921	Near-complete separation
Propensity range	[0.020, 0.935]	[0.130, 0.980]	Overlap [0.130, 0.935]
Within common support	17 of 45	32 of 122	49 of 167
Matchable within caliper	20 of 45	—	Caliper 0.527 on the logit

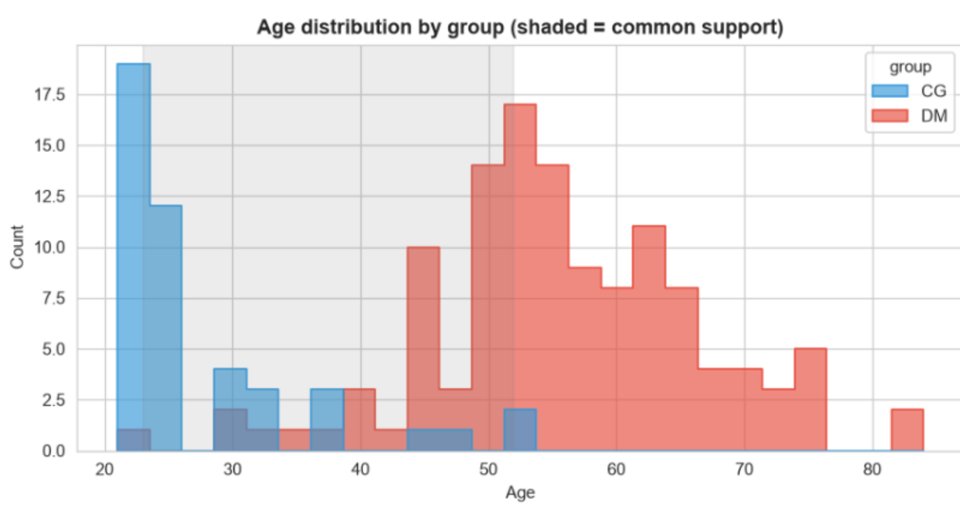

Figure 1. Age distribution by group. Controls are shown in blue and diabetic subjects in red; the shaded band marks the common-support range of 23 to 52 years. The two groups occupy largely distinct regions of the age axis.

Interpretation. The groups are close to being separate populations by age. Fewer than a third of subjects lie in the overlap region, and more than half the controls have no age-comparable diabetic counterpart anywhere in the dataset. Any covariate adjustment must therefore extrapolate substantially.

4.3 Incremental value of thermal information

Age and body mass index alone reach macro-F1 0.9178. Adding the twelve thermal features raises this to 0.9208, and adding the five biomarkers to 0.9295. Thermal features used without demographics reach only 0.7912, materially below the two-variable clinical baseline.

Table 5. Discrimination by feature block, over 50 outer folds.

Feature block	Macro-F1	95% interval	AUC
Clinical only (age, BMI)	0.9178	[0.9047, 0.9309]	0.9810
Thermal only	0.7912	[0.7729, 0.8095]	0.8920
Clinical + Thermal	0.9208	[0.9094, 0.9323]	0.9881
Clinical + Thermal + Biomarkers	0.9295	[0.9179, 0.9411]	0.9887

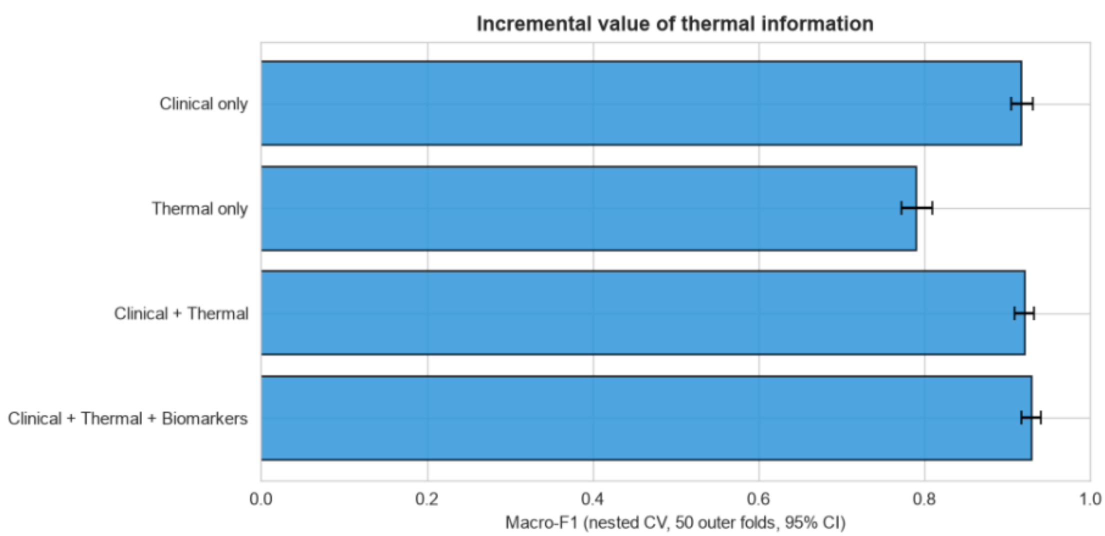

Figure 2. Macro-F1 by feature block with 95 per cent intervals. Only the thermal-only block, which is denied access to demographics, falls materially below the others.

Table 6. Paired incremental tests with the Nadeau-Bengio correction.

Comparison	Gain (pp)	Interval (pp)	p	Verdict
Clinical + Thermal vs Clinical	+0.30	[−0.86, +1.42]	0.8902	Not significant
+ Biomarkers vs Clinical + Thermal	+0.87	[+0.14, +1.56]	0.5461	Not significant
+ Biomarkers vs Clinical only	+1.17	[−0.23, +2.57]	0.6602	Not significant

The second row illustrates the point made in Section 3.4. Its bootstrap interval excludes zero while the corrected test returns p = 0.5461. The interval describes variability across folds and does not account for the dependence induced by overlapping training sets; only the corrected statistic supports an inferential claim.

Exact McNemar testing addresses the stricter question of whether thermal information changes which individuals are classified correctly. It does not: adding thermal features corrected four subjects and broke three, giving $p = 1.000$.

4.4 The share attributable to non-overlapping ages

Restricting the cohort to ages 23 to 52 leaves 79 subjects and reduces the residual age gap from 28.2 to 16.1 years. Re-executing the complete pipeline on this subset reduces macro-F1 from 0.9295 to 0.8503, a loss of 7.92 percentage points.

Table 7. Age-restricted re-analysis.

Cohort	Subjects (CG / DM)	Residual age gap	Macro-F1	95% interval
Full cohort	167 (45 / 122)	28.2 y	0.9295	[0.9179, 0.9411]
Ages 23-52 only	79 (34 / 45)	16.1 y	0.8503	[0.8221, 0.8784]
Difference	—	—	−7.92 pp	—

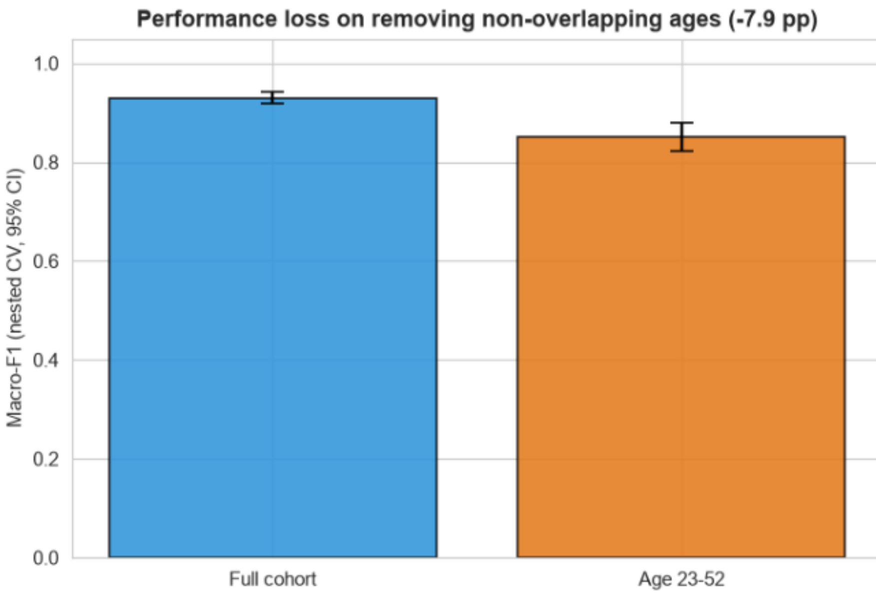

Figure 3. Discrimination on the full cohort and on the common-support band, with 95 per cent intervals. Removing the non-overlapping age range costs 7.9 percentage points.

Because 16.1 years of age difference survive inside the restricted band, this figure is a lower bound on the age-attributable share rather than an estimate of it. A genuinely age-matched cohort would be expected to show a larger loss.

4.5 Error mechanism

A classifier running on age should fail precisely where age is misleading. Of 167 subjects, nine were misclassified out of fold. Their mean age was 39.6 years against 48.9 for correctly classified subjects, a difference of 9.3 years, with Mann-Whitney $p = 0.0306$ and Cliff's delta −0.430, a medium effect.

Table 8. Direction and age profile of the misclassifications.

Error direction	n	Mean age	Group mean	Deviation
Controls predicted diabetic	4	42.0 y	27.8 y	+14.2 y, atypically old for a control
Diabetic predicted control	5	37.6 y	56.0 y	−18.4 y, atypically young for a case
All misclassified	9	39.6 y	48.9 y (correct)	−9.3 y, p = 0.0306

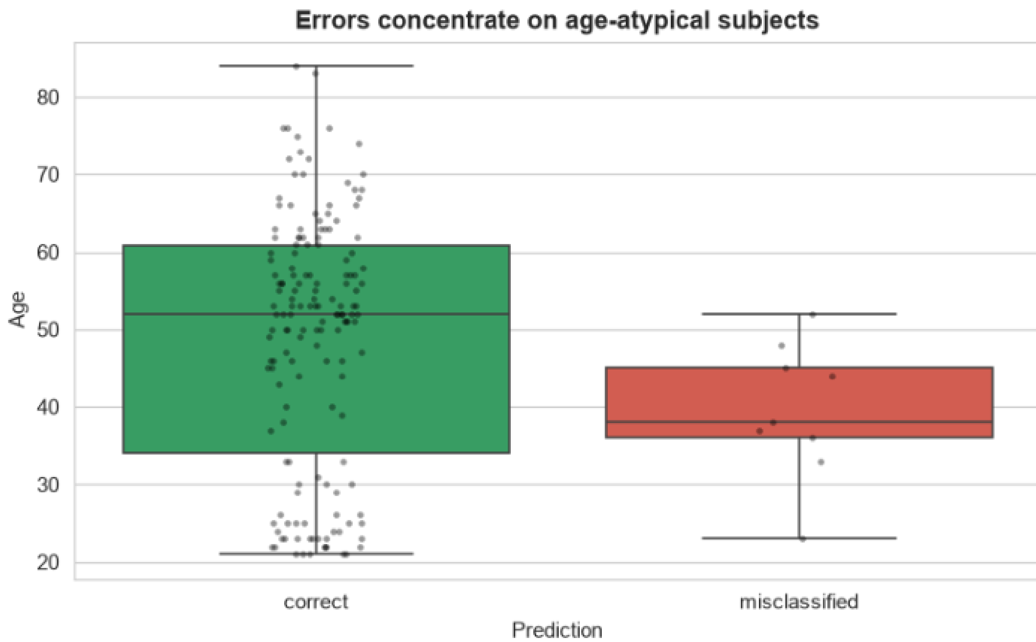

Figure 4. Age of correctly classified subjects (left, green) against misclassified subjects (right, red). Errors concentrate among younger subjects.

Interpretation. The error signature is exactly what an age-driven classifier produces: failure on older controls and younger diabetic subjects, and nowhere else systematically. Had the thermal features carried independent information, they would be expected to rescue precisely these cases. They do not.

4.6 Biomarker validation before and after adjustment

Tested without adjustment, all five biomarkers separate the groups significantly after Holm correction, with medium effect sizes for the asymmetry, heterogeneity and variability indices. Read in isolation, Table 9 would be reported as validation of the panel.

Table 9. Unadjusted biomarker comparison.

Biomarker	CG mean	DM mean	Cliff's delta	Effect	Holm p	Significant
TAI	0.366	0.766	+0.401	medium	0.00037	Yes
RHHI	10.110	10.271	+0.348	medium	0.00175	Yes
PTVI	26.193	27.070	+0.398	medium	0.00037	Yes
ASI	8.078	8.200	+0.221	small	0.01700	Yes
TSS	0.983	0.968	−0.295	small	0.00700	Yes

Adjusted for age and body mass index under Firth penalisation, none survives, in either the full cohort or the common-support subset. Every interval crosses unity and the smallest Holm-corrected p-value is 0.360. The nested penalised likelihood-ratio test for the panel beyond age and body mass index gives a chi-square of 2.403 on five degrees of freedom, p = 0.7911.

Table 10. Firth-adjusted biomarker associations, per standard deviation.

Biomarker	Full cohort OR [interval]	Holm p	Ages 23-52 OR [interval]	Holm p
TAI	3.890 [0.638, 23.738]	0.414	2.779 [0.656, 11.767]	0.360
RHHI	1.267 [0.754, 2.128]	0.414	1.311 [0.670, 2.567]	0.445
PTVI	1.357 [0.763, 2.413]	0.414	1.391 [0.697, 2.778]	0.445
ASI	1.265 [0.753, 2.125]	0.414	1.308 [0.668, 2.562]	0.445
TSS	0.439 [0.093, 2.074]	0.414	0.548 [0.152, 1.984]	0.445

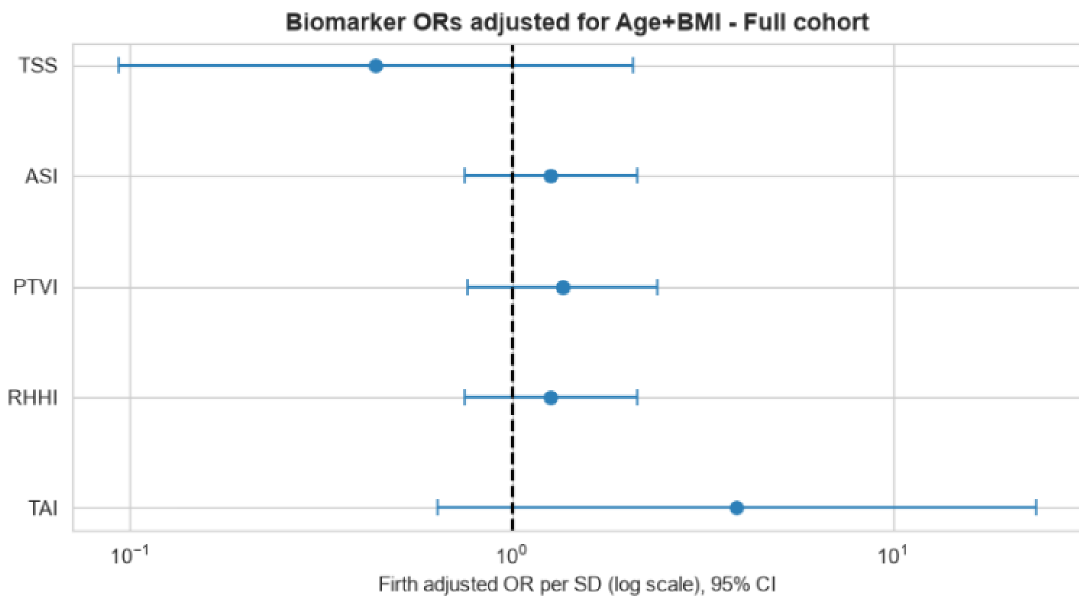

Figure 5. Firth-adjusted odds ratios per standard deviation with 95 per cent intervals, full cohort, on a logarithmic scale. Every interval crosses the dashed line at unity.

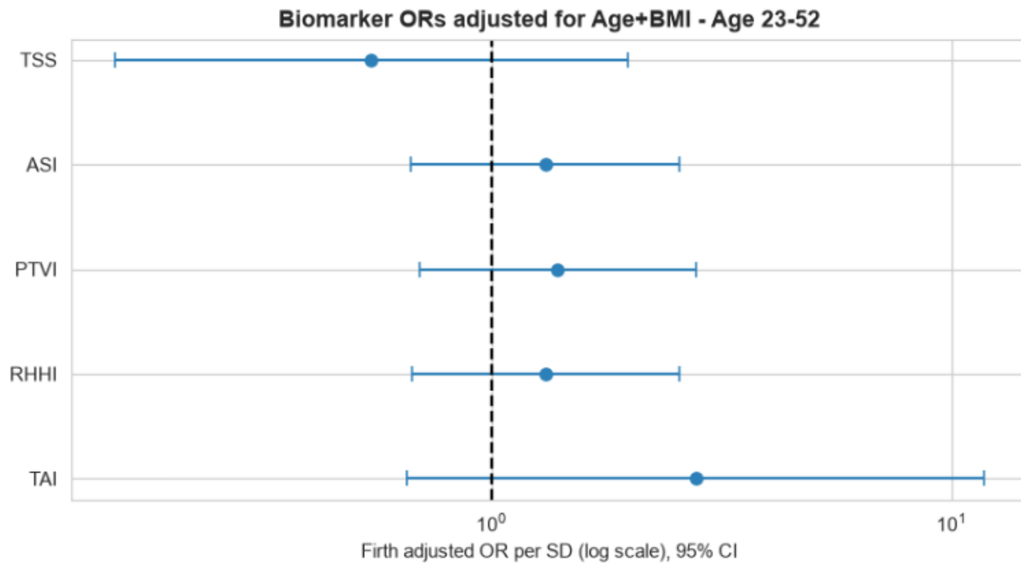

Figure 6. The same estimates within the common-support subset (n = 79). The pattern is unchanged.

The central contrast of this study. The same five biomarkers move from five of five significant to zero of five significant purely by accounting for age. A study reporting only Table 9 would draw the opposite conclusion from the same data.

One nuance should be preserved. The adjusted odds ratio for TAI is 3.890, a large point estimate whose interval is wide rather than tightly centred on unity. That is the signature of an under-powered comparison rather than of a null effect, and it anticipates the matched-cohort results of Section 4.9.

4.7 Model comparison

Algorithm choice has little influence. Macro-F1 spans 0.9177 to 0.9384 across five learners. The Friedman omnibus test is significant at $p = 0.0013$, but the pairwise comparison between the two leading models is not, at $p = 0.7206$. An omnibus difference does not license a claim that the top models differ, so both results are reported. Inner-loop threshold selection did not help at this sample size, giving 0.9200 against 0.9295 for the fixed threshold.

Table 11. Model comparison on the combined feature block.

Model	Macro-F1	95% interval	AUC	MCC	Brier
Soft Voting ensemble	0.9384	[0.9265, 0.9504]	0.9852	0.8807	0.0396
XGBoost	0.9332	[0.9217, 0.9447]	0.9892	0.8715	0.0366
Logistic Regression	0.9295	[0.9179, 0.9411]	0.9887	0.8647	0.0471
Support Vector Machine	0.9194	[0.9072, 0.9317]	0.9770	0.8435	0.0529

Model	Macro-F1	95% interval	AUC	MCC	Brier
Random Forest	0.9177	[0.9036, 0.9318]	0.9826	0.8402	0.0504

Perspective. A logistic regression on two demographic variables reaches 0.9178. A tuned ensemble on all nineteen features reaches 0.9384. Nineteen features and extensive tuning purchase approximately two percentage points over age and body mass index.

4.8 Attribution

SHAP attribution on the combined model is dominated by age, at 0.1818, more than four times the next contributor at 0.0429. Body mass index ranks third. With the clinical variables removed, attribution redistributes across the regional means, led by the lateral plantar and whole-foot averages, with asymmetry features and TAI entering the top eight.

Table 12. SHAP attribution, eight highest mean absolute values.

Rank	Combined model	Mean |SHAP|	Thermal-only model	Mean |SHAP|
1	Age	0.1818	mean_LPA	0.0577
2	mean_LPA	0.0429	mean_General	0.0536
3	BMI (IMC)	0.0377	mean_TCI	0.0443
4	mean_General	0.0359	mean_MPA	0.0438
5	mean_TCI	0.0350	asym_MCA	0.0381
6	mean_MPA	0.0346	mean_MCA	0.0366
7	mean_MCA	0.0266	asym_LPA	0.0353
8	asym_LPA	0.0219	TAI	0.0340

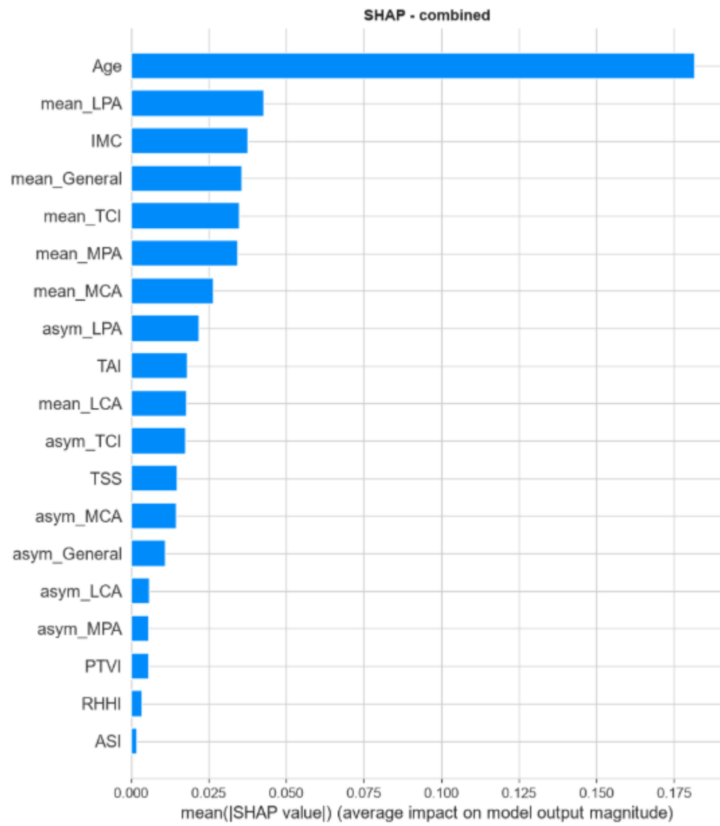

Figure 7. Mean absolute SHAP values, combined model. Age dominates all remaining features by a factor exceeding four.

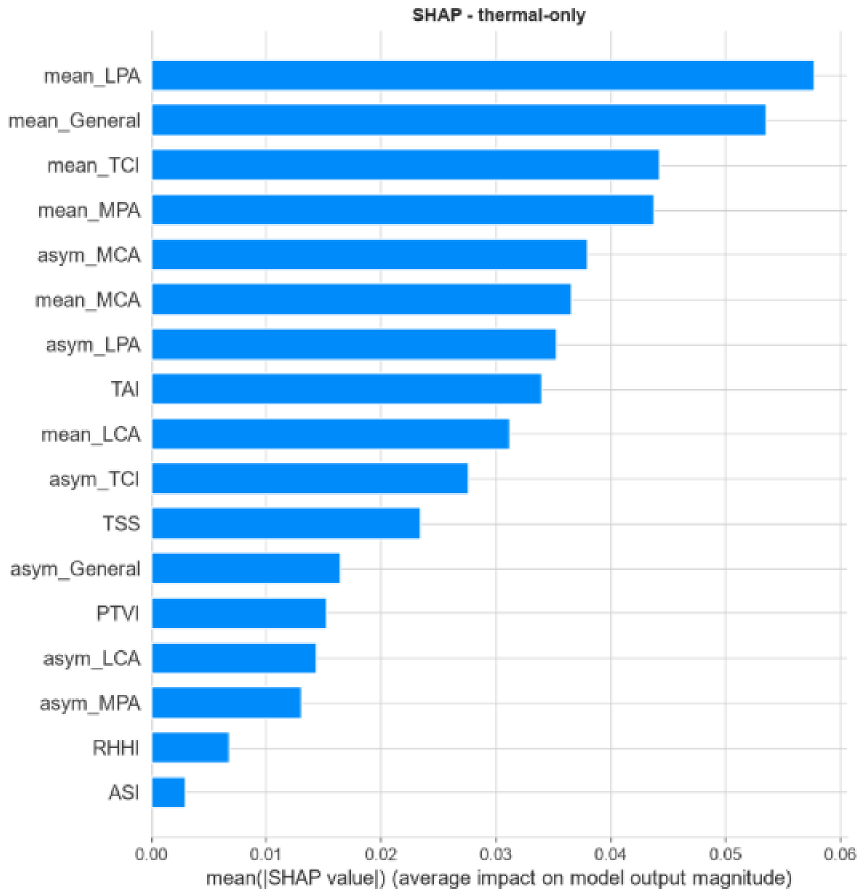

Figure 8. Mean absolute SHAP values with clinical variables removed. Attribution redistributes across regional means and asymmetry features.

That thermal features acquire attribution once age is removed shows only that the forest uses whatever remains available. It is not evidence of independent diagnostic content, and attribution describes a fitted model rather than a population.

4.9 Confounder-adjusted re-evaluation

Matching achieves its primary target and only partly the remainder. Age imbalance falls from 2.978 to −0.081 and propensity imbalance from 3.588 to 0.040, both within the conventional threshold. Body mass index remains imbalanced at 0.217, so the matched design controls age more successfully than adiposity.

Table 13. Covariate balance under each design.

Covariate	Before	After matching	Balanced	Overlap-weighted
Age	2.978	−0.081	Yes	+0.729, still imbalanced
Body mass index	0.909	+0.217	No	+0.204, still imbalanced
Propensity score	3.588	+0.040	Yes	Zero by construction

The final column deserves attention. Overlap weighting balances the propensity score exactly while leaving age imbalanced at +0.729. Results obtained under overlap weighting must not be described as age-balanced.

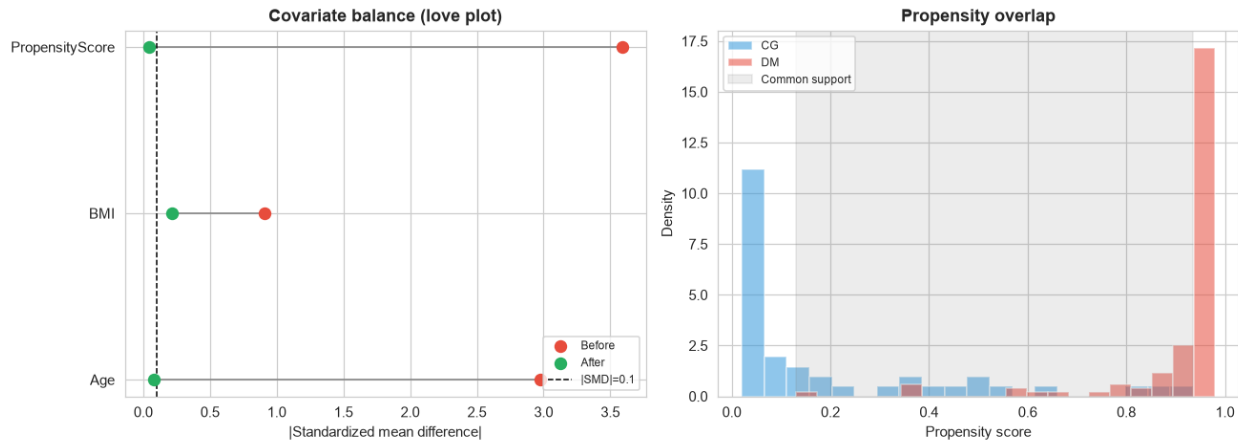

Figure 9. Balance and overlap diagnostics. Left: absolute standardised mean differences before matching (red markers) and after matching (green markers), with the 0.1 threshold marked. Right: propensity distributions, controls in blue and diabetic subjects in red, with the common-support region shaded.

Within matched pairs the picture inverts. The clinical model collapses to a median AUC of 0.488, the expected consequence of removing the variable on which it depends, while the thermal blocks retain discrimination at 0.690 and 0.730.

Table 14. Discrimination within propensity-matched sets, 1000 repetitions.

Feature block	Median macro-F1	Median AUC	Interpretation
Clinical	0.479	0.488	Indistinguishable from chance
Clinical + Thermal	0.642	0.690	Clearly above chance
Clinical + Thermal + Biomarkers	0.744	0.730	Highest of the three

Table 15. Incremental AUC under the three adjustment designs.

Design	Comparison	Gain in AUC	Interval
Repeated matching	Thermal over clinical	+0.202	[+0.157, +0.250] (resample percentiles)
Repeated matching	Biomarkers over clinical	+0.248	[+0.210, +0.281] (resample percentiles)
Overlap weighting	Thermal over clinical	+0.161	[+0.006, +0.295] (bootstrap)
Overlap weighting	Biomarkers over clinical	+0.176	[+0.009, +0.324] (bootstrap)
Stabilised IPTW	Thermal over clinical	+0.064	Not reported; unstable

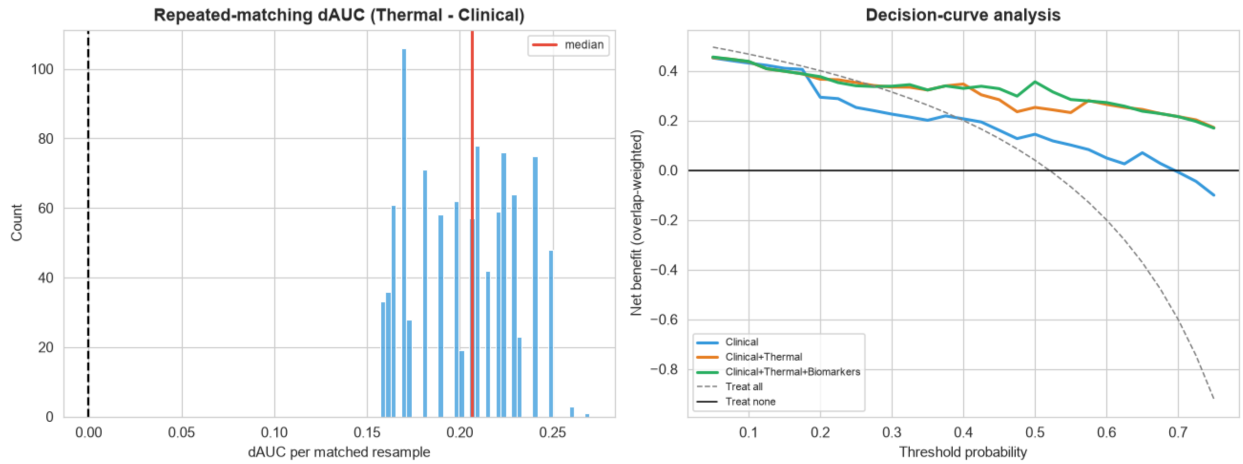

Figure 10. Left: distribution of the incremental AUC of the thermal block over the clinical block across 1000 matched resamples; the vertical line marks the median and the dashed line marks zero. Right: decision curves under overlap weights, with the clinical block in blue and the two thermal blocks in orange and green.

The paired test, however, does not move. Exact McNemar within the matched cohort of 31 subjects gives one versus four discordant pairs, $p = 0.3750$; on the full cohort, $p = 1.000$. Thermal information shifts ranking-based discrimination under the balanced designs without producing a detectable change in the classification of individual subjects.

4.10 Calibration and clinical utility

All three blocks are over-confident in the same direction on the full cohort, with calibration slopes between 1.13 and 1.20 and intercepts between 0.90 and 0.94 against ideal values of one and zero. Brier scores improve marginally as thermal information is added. On the matched cohort, absolute calibration is far worse, as expected once the easily separated age-extreme subjects are removed, but the ordering favours the thermal blocks and the gap widens.

Table 16. Calibration on the full and matched cohorts.

Feature block	Brier	Slope	Intercept	Matched Brier	Matched AUC
Clinical	0.0550	1.2045	0.9366	0.2379	0.6773
Clinical + Thermal	0.0450	1.1493	0.9290	0.1721	0.8045
Clinical + Thermal + Biomarkers	0.0438	1.1297	0.8969	0.1588	0.8318

Clinical usefulness was assessed by decision curve analysis, in which net benefit at threshold probability p is

$$NB(p) = TP/N - (FP/N) \cdot [\, p / (1 - p) \,] \qquad (10)$$

The threshold encodes the clinician's implied trade-off between a missed case and a false alarm, and a model is useful only if it exceeds both the treat-all and treat-none strategies.

Table 17. Matched-cohort net benefit by threshold probability.

Threshold	Clinical	Clinical + Thermal	+ Biomarkers	Treat all
0.10	0.2760	0.2832	0.2867	0.2832
0.30	0.0829	0.1751	0.1935	0.0783
0.50	0.0000	0.0968	0.1935	−0.2903

At a threshold probability of 0.50 the clinical model provides no net benefit at all, while the combined thermal block retains 0.1935. In decision-curve terms, the demographic model ceases to be clinically useful at the point where a decision-maker weights a missed case and a false alarm equally.

4.11 Summary of results

Table 18. Headline quantities from every stage of the analysis.

Stage	Result
Benchmark integrity	Depth-3 tree accuracy 0.856; labels not deterministically derivable
Age imbalance	28.2 years; Mann-Whitney $p = 1.52 \times 10^{-20}$; Cliff's delta +0.938
Propensity overlap	49 of 167 within common support; 20 of 45 controls matchable
Clinical-only discrimination	Macro-F1 0.9178
Thermal-only discrimination	Macro-F1 0.7912
Thermal added to clinical	+0.30 pp, p = 0.8902, not significant
Biomarkers added on top	+0.87 pp, p = 0.5461, not significant
Change in individual classifications	None detectable; McNemar p = 1.000
Cost of removing non-overlapping ages	−7.92 percentage points
Age profile of errors	9.3 years younger on average, p = 0.0306, Cliff's delta −0.430
Biomarkers significant unadjusted	5 of 5
Biomarkers significant after adjustment	0 of 5 in both cohorts
Panel beyond age and BMI	Chi-square(5) = 2.403, p = 0.7911
Clinical model within matched pairs	AUC 0.488, indistinguishable from chance
Thermal models within matched pairs	AUC 0.690 to 0.730
Incremental AUC, repeated matching	+0.202 (resample percentiles [+0.157, +0.250])
Incremental AUC, overlap weighting	+0.161 (bootstrap [+0.006, +0.295])
Paired test within matched cohort	McNemar p = 0.3750, not significant

Two distinct patterns emerge from these findings. In the benchmark as it is distributed, thermal features and thermal biomarkers do not provide any measurable advantage beyond age and body mass index: the additional gain is merely 0.30 percentage points, there are no changes in individual classifications, no biomarker withstands adjustment, and the errors are specifically concentrated on the older controls and younger diabetic subjects where age no longer offers informative value. In designs that intentionally balance age, the same predictions exhibit different behavior: the clinical model reverts to chance while the thermal blocks maintain discrimination levels between 0.690 and 0.730.

The second pattern is based on a matchable subset of twenty controls, with approximately ten pairs for each matched set and the same subjects being reused across multiple repetitions. The matched-design intervals mentioned earlier represent percentiles across match assignments rather than sampling intervals, and the paired McNemar test remains non-significant in both cohorts. Consequently, these results are presented as they are, without any claims in either direction.